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CODE:

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# Iris Flower Classification using KNN

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Iris dataset
iris = load_iris()

# Input features
X = iris.data

# Target classes
y = iris.target

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create KNN model with K = 3
knn = KNeighborsClassifier(n_neighbors=3)

# Train the model
knn.fit(X_train, y_train)

# Predict the test data
y_pred = knn.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("IRIS FLOWER CLASSIFICATION USING KNN")
print("=" * 45)

print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))

# Classify a new flower
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = knn.predict(new_flower)
```

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print("\nNEW FLOWER DETAILS")
print("-" * 45)
print("Sepal Length:", new_flower[0][0])
print("Sepal Width :", new_flower[0][1])
print("Petal Length:", new_flower[0][2])
print("Petal Width :", new_flower[0][3])

print("\nPredicted Flower:", iris.target_names[prediction[0]])
```

OUTPUT :

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      setosa      1.00      1.00      1.00      10
versicolor      1.00      1.00      1.00       9
 virginica      1.00      1.00      1.00      11

      accuracy
macro avg      1.00      1.00      1.00      30
weighted avg      1.00      1.00      1.00      30

NEW FLOWER DETAILS
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Sepal Length: 5.1
Sepal Width : 3.5
Petal Length: 1.4
Petal Width : 0.2
```